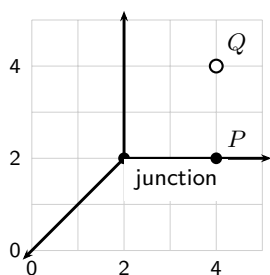


For a point (x, y) , compare x , y , and 2. When the least number occurs at least twice, the point lies on this line.



$$P = (4, 2) : 4, \underline{2}, \underline{2}$$

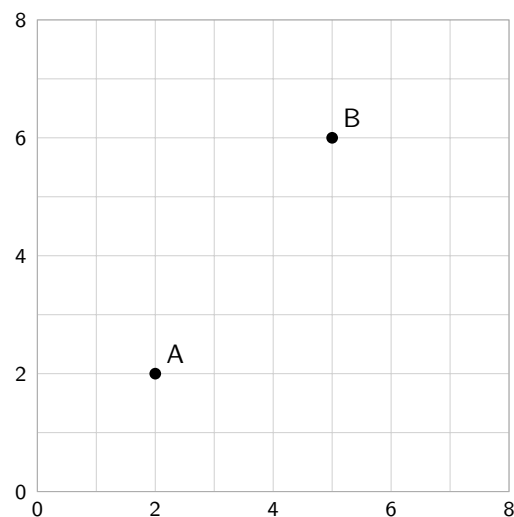
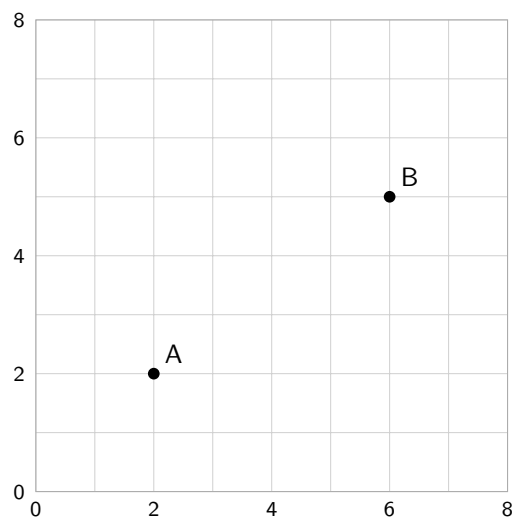
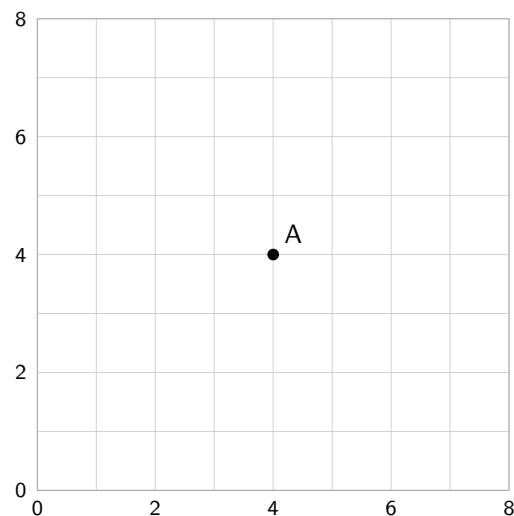
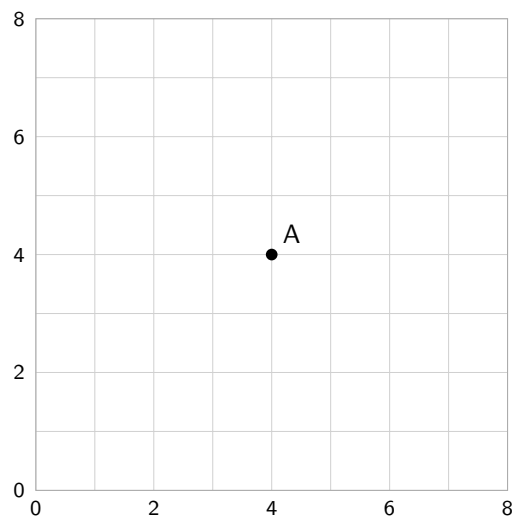
The least number is tied. P is on the line.

$$Q = (4, 4) : 4, 4, \underline{2}$$

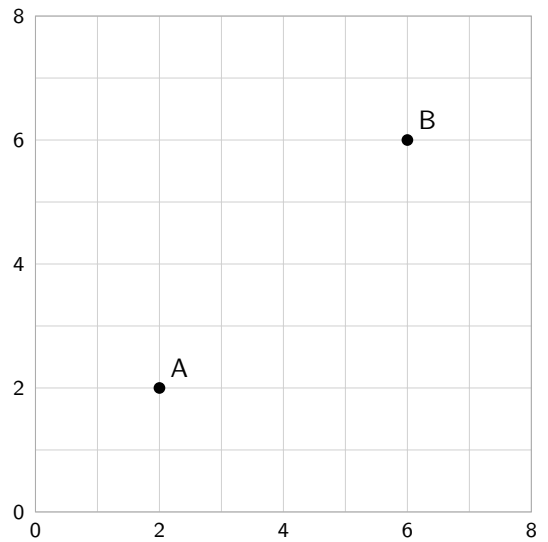
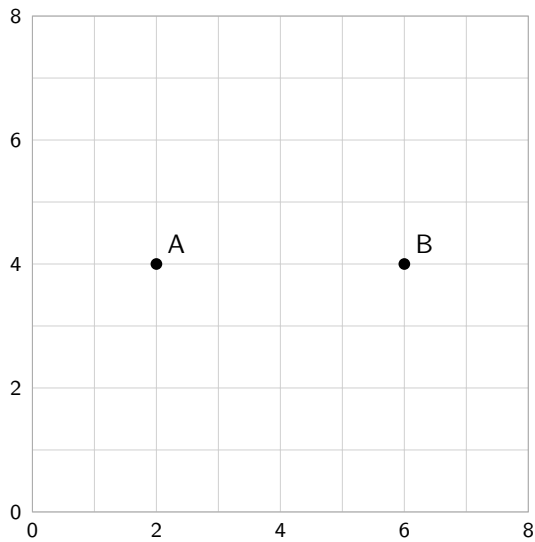
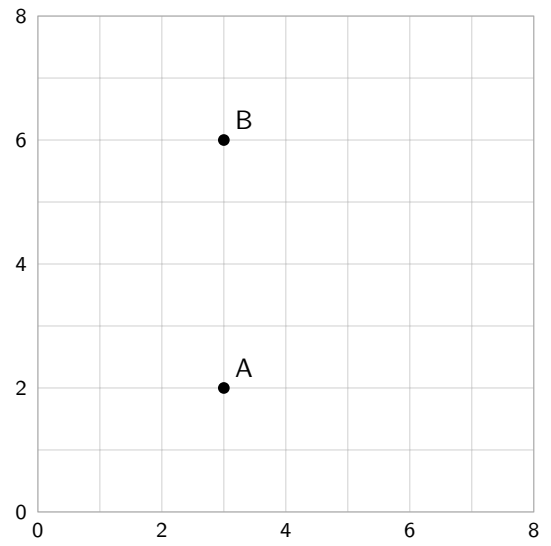
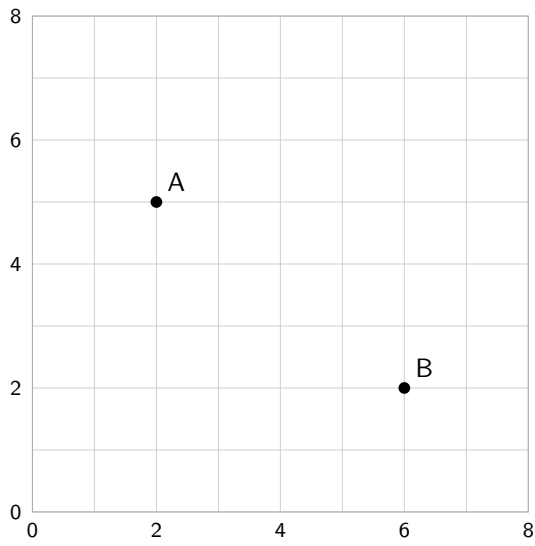
The least number occurs once. Q is off the line.

The arms point east, north, and diagonally southwest. They continue without end. Every line in this packet has this shape, moved without turning. Points between grid dots count too.

Problem 1: Choose junction B on each top grid. Draw the lines with junctions A and B in different colours on every grid. Make your chosen pairs meet in different ways. Circle single shared points and trace shared stretches heavily.

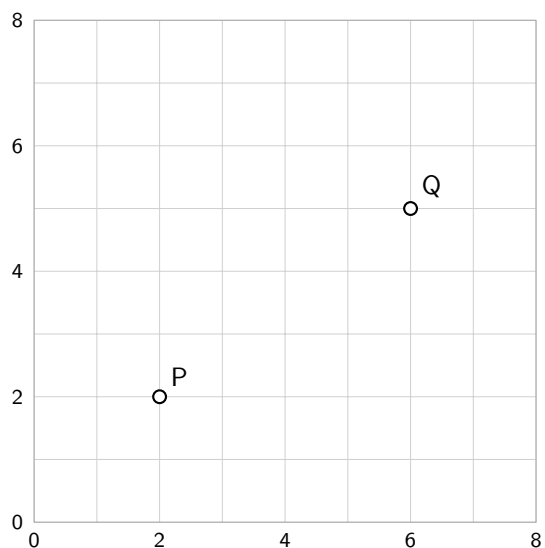
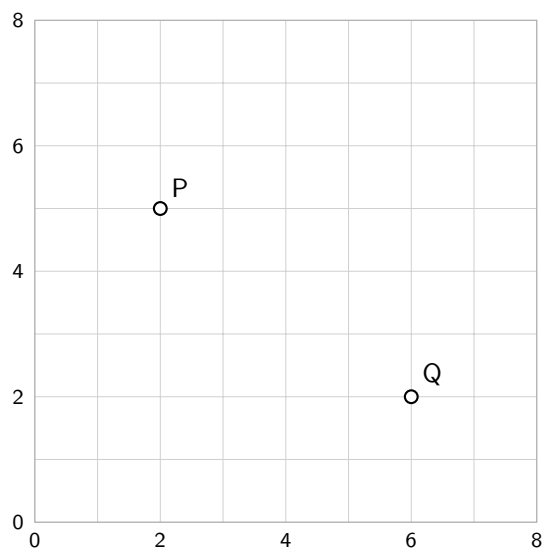


Problem 2: Draw the lines with junctions A and B on each grid. Mark every part they share. How many shared points does each pair have?

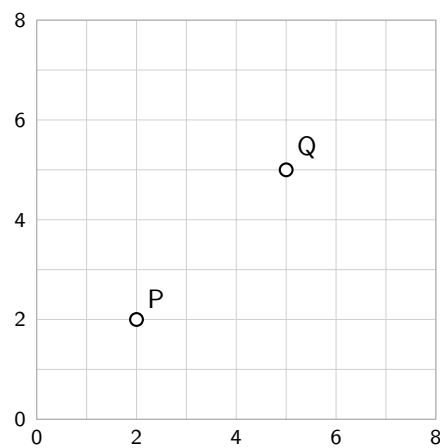
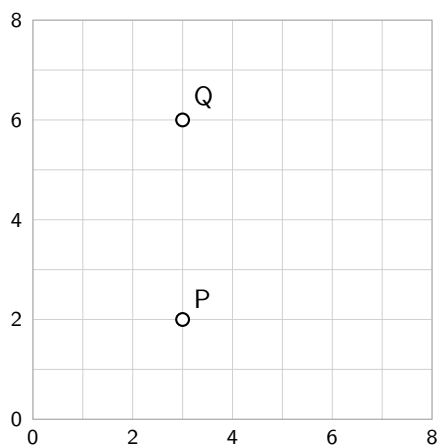
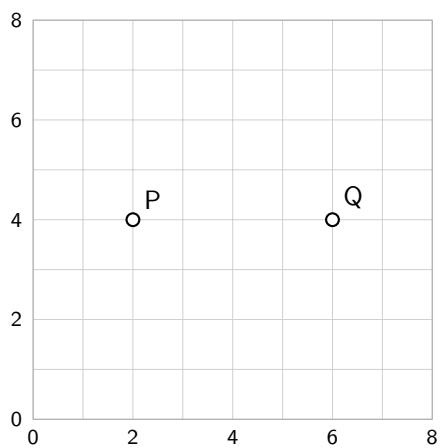


Problem 3: Can two lines with different junctions share exactly two points? Draw an example, or explain why this cannot happen.

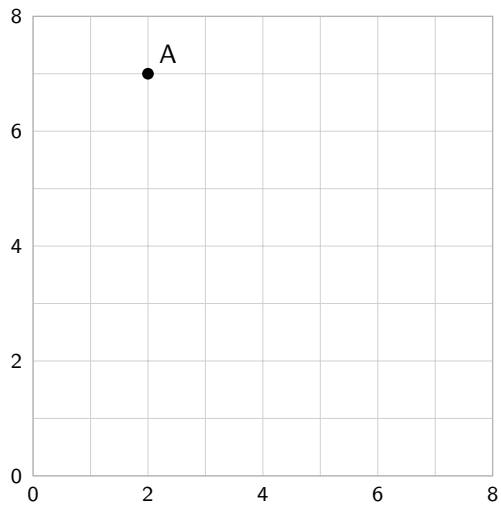
Problem 4: Draw a line through both points on each grid. Its junction may be anywhere, including between grid dots or outside the grid. Can you find a second line through the same two points?



Problem 5: Find all possible junctions of a line through each pair of points. Mark a whole stretch if every point on it works.



Problem 6: Find where the lines with junctions $A = (2, 7)$ and $B = (10, 5)$ meet. Can any two lines miss each other? Explain your answer. Give a rule for when two lines share more than one point. Your rule must also work for junctions between grid dots or outside this page.



Problem 7: Find a rule that tells how many lines can pass through two different points. Explain why your rule works.

